School of Computing and Mathematics sciences

CO7201 Individual Project

Preliminary Report

Placement management web application

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Placement management system

1. Introduction:

Universities conduct campus placement drives to help students find jobs. However, many universities and colleges still handle placement activities through manual or partly digital methods like emails, spreadsheets, and informal communication. These approaches have several problems, including inefficiency, data errors, scheduling issues, and a lack of transparency in application statuses.

This system looks at the potential of using a placement management system to improve the efficiency and effectiveness of campus placements. The project highlights the real-world challenges institutions face and aims to enhance placement outcomes and administrative efficiency. This project explains how the placement management system can boost the efficiency and effectiveness of the recruitment process compared to regular manual and semi-digital methods.

Placement Management System is an effective business solution for streamlining and automating the complete campus recruitment process. The core modules consist of user authentication that can distribute secure access within students, placement officers, companies and admin; student profile management for updating personal academic information along with skills and resumes; real-time notification system to signal important events such as deadlines or interviews without refreshing the page, and robust database management for proper storage.

The Recommended modules add on to make the system more intelligent by providing an analytics dashboard for monitoring statistics for placement, a document verification module that validates eligibility and certificates uploaded, and an automated scheduling system that avoids clashes in interviews. Optional modules provide advanced features with a recommendation system to connect students with jobs based on skills, machine learning

providing success/probability of placement predictions, and Google Maps capabilities to visualize company locations and those that have been placed.

1. 2. Aims and Objectives:

Aims:

The main goal of this project is to create a web-based placement management system that improves the effectiveness, efficiency, and transparency of the campus hiring process. The digital platform will assist students, placement officers, and recruiters at all stages of placement, from job posting to application, interview scheduling, and managing job offers. This online platform will replace the current manual process. It will provide users with access to placement-related data and reduce administrative work while improving data accuracy.

- It enhances the accessibility of relevant job opportunities by enabling students to search and apply for relevant job positions that are available in the centralized portal.
- Providing career counseling and mentoring support through communication between students and placement coordinators to help students align their skills and with the requirements of the company
- Improves readiness of students by providing training programs, workshops, technical seminars, and mock interviews on the platform, thereby helping in skill development.
- Institutional visits & training for hands-on experience in the industry.
- There, students may practice and develop professional competencies that are critical in workplaces impacting their preparedness to enter the working world, such as workplace and communication skills.
- Facilitating placement officers with high management controls to plan interviews, monitor the positions of applications and design reports that contribute towards decision making related data collection.

Objectives:

- To develop the modules for students, placement officers, tutors, and Administrators.

- To provide a centralized management for job posting, applications, interview schedules, and offers.
- automated reminders for deadlines, interviews, and placement updates
- To provide support through mock interviews, feedback, and resource sharing.
- To develop Reporting and analytics dashboards for placement performance tracking.
- Evaluating the system based on Usability, performance, reliability and user satisfaction

1.3. Challenges

- In developing the web-based placement management system, we face several technical and operational challenges.
- We address these challenges carefully to improve the system's effectiveness, scalability, and reliability.
- This project involves multiple roles, including students, placement officers, companies, and administrators; therefore, system integration and design are essential.
- Handling large volumes of student profiles, resumes, company information, and other data requires a well-structured database design.
- Processing real-time notifications, such as interview alerts and deadline reminders, without impacting system performance presents a challenge.
- The system manages personal information for students and companies, including resumes and contact details, which raises security and privacy concerns.
- We must prioritize operational challenges like user adoption, change management, scalability, and performance.

2. Requirements

Priority	Requirement	Purpose/Description
Essentials	User authentication	Securing logins and registrations for students, placement, officers, companies, administrators.
	Student Profile Management	Creating and updating profiles, skills, and resumes.
	Notification System	Sending notifications for deadlines and interviews, without any page refreshes
	Database Management	Storing placement-related information while maintaining integrity and consistency.
Recommended System	Analytics Dashboard	Analysing data such as placement percentages, company hiring rates and student performances.
	Document Verification Module	Verifying Uploaded documents and checking the eligibility
	Automated Scheduling conflict	Preventing Overlapping of interviews.
Optional System	Recommendation System	Suggesting jobs based on student's skills and performance.
	Placement Prediction Model	Predicting the probability of placement success.
	Google Maps Integration	Displaying company locations and students who have been placed.

3. Technical Specification

The placement management system will be built using modern technologies. The project has a three-tier structure that includes the presentation layer, the user application layer, and the data layer.

We will use frontend technologies like React.js (HTML, CSS, JavaScript) to create an interface for students, placement officers, companies, and administrators. For the backend, we will use Spring Boot to handle business logic, the interview process, applications, and reporting for job postings. MySQL will serve as the database to store structured data.

For authentication and authorization, we will implement JWT (JSON Web Tokens) to secure passwords and the encryption process. Git will manage version control during the system's development. To check reliability, performance, and usability, we will carry out integration testing, unit testing, and functional testing.

4. Background Research

Candidate skills increased by web-based recruitment from automating recruitment process to improving hiring efficiency Securing authentication details are must. application tracking module is influenced. So digital platforms improve the transparency and accessibility.[1]

And integrated systems reduce communication gaps; feedback modules will increase usability and accessibility. [2]

Data Mismanagement can happen in manual placement processes which can also lead to scheduling conflicts, automation in tracking, reducing workload in online systems and by designing a structured database. [3]

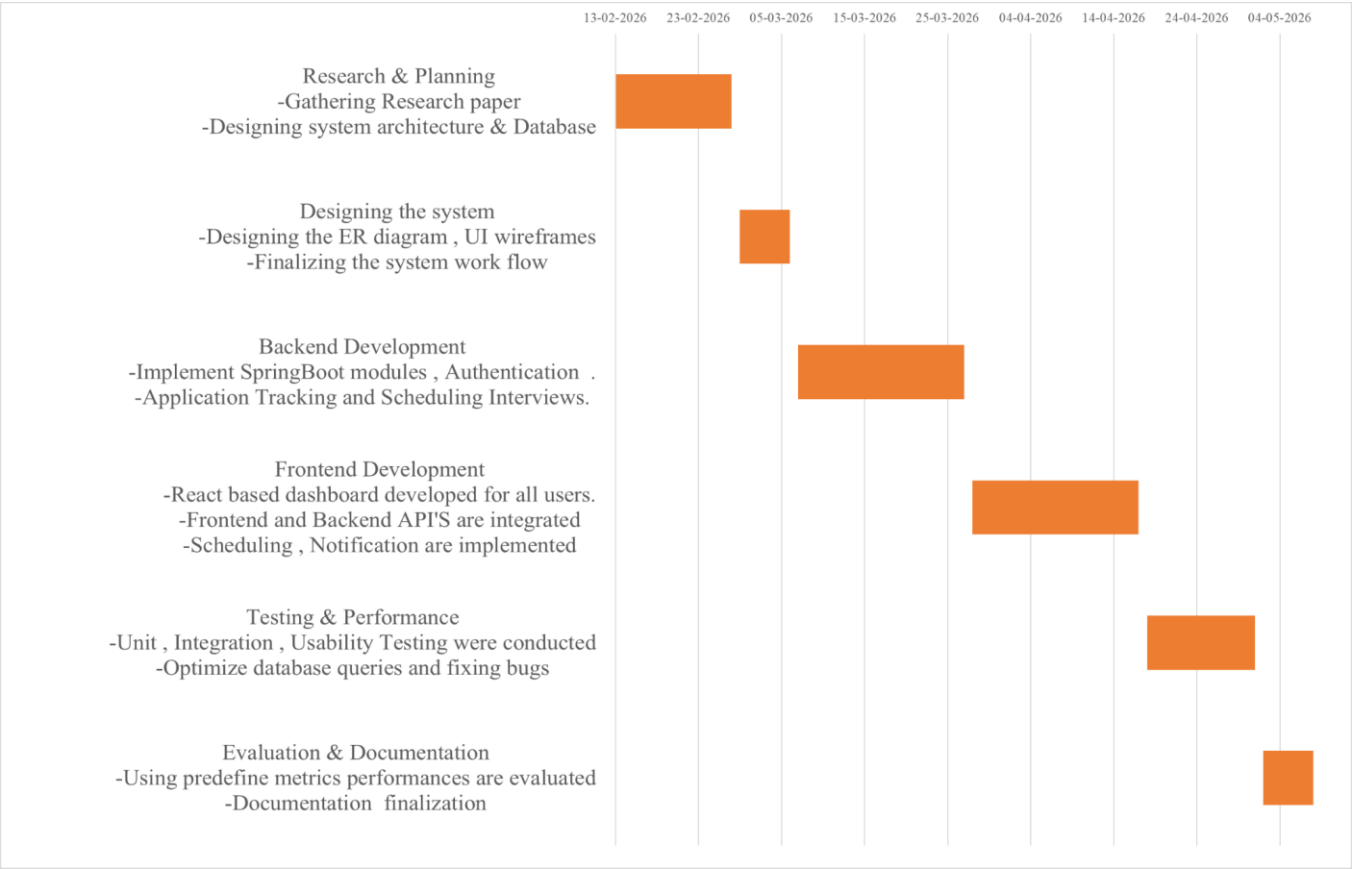
Easing the efficiency of job offers based on keyword matching. It requires structured data of the student profile. [4]

For predicting placement outcomes based on academic performance machine learning models are used. are inspired from the reporting and placement performance tracking. You are data up to October 2023 for ML storing. [5].

Solution: Placement Management System A placement management system is an application software that provides university placement coordinators with the tools needed to manage and access information about placements[6].

There are many systems out there which have data redundancy, no real-time tracking as discussed in this research identifying the need of eligibility-filtering and secured access control. [7]

5.1 Time Plan



5.2 Risk Plan

Risk	Probability	impact	Mitigation Strategy
Backend development delays	Medium	High	Develop core modules first, follow an incremental approach.
Scheduling conflicts can be complex.	Medium	Medium	Design the algorithm first and test with sample data.
Failures in real-time notifications	Medium	Medium	Provide email alerts and maintain logs for debugging.
Performance issues during placement time	Medium	High	Implement indexing and optimize queries.
Lack of progress tracking	Low	Medium	Use Git version control with regular commits and monitor progress.
Loss of data	Low	High	Backup regular databases
Difficulty in measuring system effectiveness	Medium	Medium	Usability scores, error rates, and response times.

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